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RESEARCH INTERESTS

Physics of gamma-ray burst afterglows — structured jets and off-axis geometry; short GRBs and energetic long GRBs; both prompt-emission and afterglow modeling; GRB population studies; electromagnetic counterparts of compact-object mergers; fast transients in time-domain surveys and rapid follow-up.

Methods: image subtraction and machine-learning transient detection; end-to-end automation of alert-driven follow-up — telescope operations, data reduction, and physical modeling as one chain; Bayesian inference, model comparison, and the model systematics that limit afterglow parameters.

EDUCATION

Ph.D. in Physics (Astronomy & Astrophysics) , Indian Institute of Technology Bombay <i>Study of Electromagnetic Emission from Compact Objects</i> · Advisor: Prof. Varun Bhalerao	2021–current
M.Sc. Physics , Department of Physics & Astrophysics, University of Delhi	2018–2020
B.Sc. (Hons.) Physics , Deen Dayal Upadhyaya College, University of Delhi	2015–2018

PUBLICATIONS

21 refereed publications (2 first-author, 1 second-author; co-author on one Nature paper); h-index 11. Complete list: [SciX](#).

First author

1. Swain, V., et al. 2026, *GRB 250704B: An Off-axis Short GRB with a Long-lived Afterglow Plateau*, [ApJL](#), 996, L38
A rare case where the afterglow shape is set by viewing geometry rather than energetics or environment: a cosmological analog of GW170817, and a template for interpreting the next gravitational-wave counterpart.
2. Swain, V., et al. 2026, *GRB 230204B: GIT Discovery of a Fast Fading Afterglow Associated with an Energetic Gamma-Ray Burst from a Massive Star Progenitor*, [ApJ](#), 1000, 58
Discovery and complete characterization of one of the most energetic bursts known, from prompt gamma-rays through a rapidly fading afterglow to its Wolf–Rayet progenitor wind.

In preparation

3. Swain, V., et al., *The GIT transient-detection pipeline: image subtraction and machine-learning candidate classification* — in preparation, arXiv September 2026
4. Swain, V., et al., *A homogeneously modeled catalog of GIT-observed GRB afterglows* — in preparation

Selected co-author (role in brackets)

- Kasliwal, M. M., et al. 2025, *AT 2025ulz and S250818k: A Candidate Superkilonova from a Subthreshold Subsolar Gravitational-wave Trigger*, [ApJL](#), 995, L59 [GRB afterglow modeling, HCT follow-up, data analysis, and interpretation of the counterpart as an off-axis structured jet]
- Ho, A. Y. Q., et al. 2023, *Minutes-duration optical flares with supernova luminosities*, [Nature](#), 623, 927 [GIT follow-up and special handling of clear-filter data with colour terms, fringe correction, and cross-telescope image subtraction]
- Ahumada, T., et al. 2026, *S250206dm: Zwicky Transient Facility Observations*, [PASP](#), 138, 034101 [candidate scanning in ZTF; GIT follow-up to confirm optical afterglows]
- Eappachen, D., et al. 2026, *Characterizing EP241107a: an Einstein Probe-detected fast X-ray transient*, [MNRAS](#), 545 [GIT follow-up; GRB-EP connection and interpretations]
- Srinivasaragavan, G. P., Swain, V., et al. 2024, *Characterizing the Ordinary Broad-line Type Ic SN 2023pel from the Energetic GRB 230812B*, [ApJL](#), 960, L18 [second author — GIT follow-up; image subtraction across multiple telescopes; GRB interpretation]
- Li, M. L., et al. 2025, *The Nature of Optical Afterglows without Gamma-Ray Bursts: Identification of AT2023lcr*, [ApJ](#), 985, 124 [discovered AT2023lcr while scanning the ZTFReST stream; GIT follow-up]
- Srinivasaragavan, G. P., et al. 2025, *Multiwavelength analysis of AT 2023sva: a luminous orphan afterglow with evidence for a structured jet*, [MNRAS](#), 538, 351 [GIT follow-up]
- Ahumada, T., et al. 2024, *Searching for Gravitational Wave Optical Counterparts with the Zwicky Transient Facility: Summary of 04a*, [PASP](#), 136, 114201 [counterpart scanning; GIT follow-up]
- Kumar, H., et al. 2022, *India's First Robotic Eye for Time-domain Astrophysics: The GROWTH-India Telescope*, [AJ](#), 164, 90 [GIT operations handover; filter colour terms]

Plus 10 further refereed publications on supernovae, novae, solar-system objects, and exoplanet ephemerides — listed in full at the end of this CV.

Rapid communications

- 130 GCN Circulars (20+ first-author, including the GIT afterglow discovery circulars for GRB 230204B and GRB 251126A)
- 20 Astronomer's Telegrams

- 18 TNS discovery reports from routine scanning of ZTF transient candidates
- 100+ Minor Planet Electronic Circulars (MPECs) for near-Earth objects and comets

RESEARCH & OBSERVATIONAL EXPERIENCE

GROWTH-India Telescope (GIT), 0.7 m, Hanle

2021–2026

- Core responsibility for GIT operations and maintenance 2022–2026
- GRB and Einstein Probe follow-up throughout; discovered the optical afterglows of **GRB 230204B** and **GRB 251126A**.
- **Team leader, LVK O4 follow-up operations (2023–2025)** – built and ran the end-to-end response system; triggered on S230615az, S231113bw, and S241125n.
- Near-Earth object follow-up (2021–2024): 100+ objects observed with GIT, contributing to published MPECs.
- **GIT on-site annual maintenance, Indian Astronomical Observatory, Hanle (2021)** – two-week visit working hands-on with the GIT hardware alongside observatory engineers and observers.

Zwicky Transient Facility – ZTFReST and LSST-FAST scanning

- Regular scanning for fast transients; **discovered the orphan afterglow AT2023lcr** (ZTF23aaooohpy), published in Li et al. 2025.
- Further discoveries: ZTF23aaxzvrr (Galactic nova), ZTF23abbfrqp (flaring candidate); 18 TNS discovery reports.
- ZTF and DECam scanning for optical counterparts of gravitational-wave events and short GRBs.
- Routine scanning for near-Earth objects in ZTF data (ZStreaks, 2021–2024).

Target-of-opportunity and DDT observing

Triggered follow-up on HCT, SEDM, NGPS, *Swift*, GMRT, and *Chandra*.

TELESCOPE PROPOSALS

- **PI / Co-I on 21 accepted Himalayan Chandra Telescope proposals** – follow-up of gamma-ray bursts, Einstein Probe transients, and electromagnetic counterparts of gravitational-wave events, including the LIGO IR1 observing cycle.
- **DDT: Chandra X-ray follow-up of GRB 230812B and GRB 250704B; GMRT radio follow-up of GRB 260516D.**

SOFTWARE & PIPELINE DEVELOPMENT

GRB analysis

- End-to-end afterglow analysis pipeline: data ingestion, extinction correction, empirical fitting, physical modeling (nested sampling, evidence/BIC comparison), cross-code systematics validation, degeneracy and parameter-correlation analysis, population statistics, and publication-ready output.
- Prompt-emission pipeline for time-integrated and time-resolved analysis of *Fermi*-GBM and *Swift*-XRT data.

Survey and transient detection

- GIT transient-detection system: source-injection training-set generation, image subtraction, statistically validated analytic vetting, ML-based real/bogus classification, and a human-scanning interface.
- Rebuilt the automated nightly pipeline: reduction, recovery from failed astrometry, saturation and bad-pixel masking, fringe correction, PSF photometry with weight maps, image subtraction, median and RMS-weighted stacking, analytic and ML vetting, and a nightly scanning interface.

Observatory operations

- Fully automated LVK O4 follow-up system (as team leader): alert and skymap ingestion → tiling → scheduling → automated or manual triggering → observation and data transfer → reduction → human scanning → GCN publication.
- GIT operations improvements: ToO scheduling algorithm and terminal interface, filter-wheel error handling, weather-alert system, nightly reporting, and multi-institute data synchronization.

TECHNICAL SKILLS

- **Optical/NIR data reduction:** GIT, HCT (optical and NIR), ZTF, LT, SEDM, LDT
- **High-energy data reduction:** *Fermi*-GBM, *Swift*-XRT
- **Modeling frameworks:** afterglowpy, jetsimpy, redback, VegasAfterglow, 3ML, Fermi GBM Data Tools, EXOTIC
- **Analysis stack:** Astropy, Astroplan, Photutils, Astro-SCRAPPY, SExtractor, SEP, PSFEx, SCAMP, SWarp, astrometry.net (solve-field)
- **Inference and ML:** emcee, PyMultiNest, UltraNest, nautilus, PyTorch, scikit-learn
- **Computing:** Python, Linux, Git/GitHub, SQL, $\LaTeX$, HTML

AWARDS & HONORS

- **ASI Zubin Kambhavi Award**, Astronomical Society of India, 2026 – team award to the GROWTH-India Telescope team
- **Dhananjay Joshi Fundamental Research Award**, 2026 – team award to the GROWTH-India Telescope team
- Best presentation, live demonstration of the GIT robotic system, Amateur Astronomy Conference, IIT Bombay, 2023

- GATE Scholarship, Government of India, 2021 (doctoral studies); All India Rank 224, GATE Physics 2021; Rank 13, University of Delhi M.Sc. entrance examination

TALKS & PRESENTATIONS

Invited

- *A Treasure Hunt in the Sky: Hands-on Exploration of Optical and NIR Archival Surveys* – public talk, Wilson College (Autonomous), Mumbai 2025

Contributed talks

- *GRB 250704B: An Off-axis Short GRB with a Long-lived Afterglow Plateau* – Young Astronomers' Meet 2025
- *GRB 250704B* – SymPhy national conference, IIT Bombay 2025
- *GRB 230204B: A Highly Energetic and Fast Fading GRB* – 42nd Astronomical Society of India (ASI) Meeting 2024
- Live demonstration of the GIT robotic system – Amateur Astronomy Conference, IIT Bombay 2023
- Regular talks at the IITB Astronomy Journal Club and the Time-Domain Astronomy Club (IIT Bombay, TIFR, IIA, IUCAA)

Posters

- *Rapid follow-up of gravitational wave events with a robotic meter-class optical telescope* – IAU General Assembly XXXII, Cape Town 2024
- *Rapid follow-up of gravitational wave events with a robotic meter-class optical telescope* – 43rd Astronomical Society of India (ASI) 2025
- *Studying Near-Earth Asteroids with the wide-field GROWTH-India Telescope* – 41st ASI Meeting 2023
- *Solar System Studies with the GROWTH-India Telescope* – 40th ASI Meeting 2022
- *Solar System Studies with the GROWTH-India Telescope* – Young Astronomers' Meet 2022

TEACHING & MENTORING

Teaching assistant, IIT Bombay – Electronics Laboratory (three semesters, 2022–23), Astronomy & Astrophysics (2023–24), Undergraduate Physics Laboratory (2024–25)

Research mentoring – supervised 20+ undergraduate and junior graduate students (2021–present) on projects spanning telescope operations, pipeline development, and GRB science:

- *Observations and operations*: GIT observing and robotic operations, automated focus loops, new camera installation and automation, dashboard and database interfaces – S. Samantaray, M. Gulati, K. Adithya Krishna, R. Sharma, Ravi Kumar, T. Mohan
- *Transient detection*: image subtraction, source injection for training-set generation, ML real/bogus classification, scanning interfaces – A. Suresh, D. Raman, S. Gholap, M. Goyal, A. Lonakadi, V. Vijaykumar
- *GRB science*: multiband and broadband afterglow modeling, catalog data assembly, *Fermi*-GBM prompt analysis – Y. Wagh, M. Goyal, Aryan Kumar, H. Tanenia, Athul, S. Patil
- *Gravitational-wave follow-up*: GIT tiling of LVK skymaps – H. Choudhary, T. Roychowdhury
- *Solar system*: near-Earth object observing, ZTF streak scanning, Astreaks pipeline development – H. Choudhary, Y. Wagh, S. Tiwari, V. Vijaykumar
- Several mentees are now first authors on GIT publications and circulars; two are continuing my GIT operations and afterglow-modeling work.

Krittika Summer Projects 5.0 (two-month program, 2024) – *Study of the Nearest Hydrogen-Rich Core-Collapse Supernova: SN 2023ixf*

Outreach mentoring – supervised 9 school and first-year students (2023–2026) on short research projects in GRB physics and optical astronomy.

COLLABORATIONS

GROWTH-India – operations, observing, and pipeline development • **ZTF (ZTFReST)** – fast-transient scanning • **GROWTH-MMA** – ZTF and DECam scanning for gravitational-wave and short-GRB counterparts • **LSST-FAST** – fast-transient searches with Vera C. Rubin Observatory

SERVICE & OUTREACH

- Volunteer, *Transients 2024* conference and the SERB National Research meeting, IIT Bombay
- Public demonstrations of LIGO, EMGW, and Daksha satellite science at IIT Bombay Techfest (2021–2026) and VJTI Mumbai (2023)
- Regular live demonstrations of GIT operations for laboratory visitors
- Volunteer educator, Padhao Campaign (Rann Bhoomi Foundation) and NSS, University of Delhi (2016–17)

REFERENCES

- **Prof. Varun Bhalariao** – IIT Bombay (PhD advisor) – varunb@iitb.ac.in

FURTHER REFEREED PUBLICATIONS

Remaining refereed publications, reverse chronological. Author order, journal, volume, and page verified against the published record.

1. Chamoli, S., et al. 2025, *Challenging Classical Paradigms: Recurrent Nova M31N 2017-01e, a BeWD System in M31?*, **ApJ**, 991, 174
2. Basu, J., et al. 2025, *Discovery and Detailed Study of the M31 Classical Nova AT 2023tkw: Evidence for Internal Shocks*, **ApJ**, 980, 129
3. Singh, A., et al. 2024, *Unravelling the Asphericities in the Explosion and Multifaceted Circumstellar Matter of SN 2023ixf*, **ApJ**, 975, 132
4. Teja, R. S., et al. 2024, *SN 2021www: A Core-collapse Supernova at the Subluminous, Slower, and Shorter End of Type IIPs*, **ApJ**, 974, 44
5. Basu, J., et al. 2024, *Multiwavelength Observations of Multiple Eruptions of the Recurrent Nova M31N 2008-12a*, **ApJ**, 966, 44
6. Sicardy, B., et al. 2024, *Constraints on the evolution of the Triton atmosphere from occultations: 1989–2022*, **A&A**, 682, L24
7. Teja, R. S., et al. 2023, *Far-ultraviolet to Near-infrared Observations of SN 2023ixf: A High-energy Explosion Engulfed in Complex Circumstellar Material*, **ApJL**, 954, L12
8. Srinivasaragavan, G. P., et al. 2023, *A Sensitive Search for Supernova Emission Associated with the Extremely Energetic and Nearby GRB 221009A*, **ApJL**, 949, L39
9. Sharma, K., et al. 2023, *Astreaks: astrometry of NEOs with trailed background stars*, **MNRAS**, 524, 2651
10. Pearson, K. A., et al. 2022, *Utilizing a Global Network of Telescopes to Update the Ephemeris for the Highly Eccentric Planet HD 80606 b and to Ensure the Efficient Scheduling of JWST*, **AJ**, 164, 178